# final

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## 1 Analítica de Datos Proyecto Final

## 2 Regresión Logística

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Usando el conjunto de datos siguiente: https://www.kaggle.com/brllrb/uberand-lyft-dataset-boston-ma, construir un modelo de regresión logística binario. El objetivo es encontrar patrones entre las solicitudes realizadas a Uber y Lyft. También se hará un análisis de los resultados usando como métricas: la exactitud (accuracy), la sensibilidad (recall) y la métrica F1 (Score F1). Regresión logistica: Solicitudes Uber & Lyft

## 3 Contenido

1. Preparar los datos
2. Análisis exploratorio de datos
3. Visualizar los datos
4. Crear la matriz de atributos y el vector de etiquetas
5. Escalar los datos (para cada atributo)
6. Evaluar el error
7. Realizar la regresión
8. Comparar las predicciones

## 4 Importamos Librerías

[92]:
```
import os
import zipfile
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, MinMaxScaler
```

```
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix
```

## 5 Asignamos la ruta del dataset

[93]:
```
PATH = "datasets"
```

## 6 1. Preparación de los datos

[94]:
```
with zipfile.ZipFile(PATH + '/archivo.zip', 'r') as archive:
    # Extrae todo el contenido del archivo zip en el directorio actual
    archive.extractall(PATH)
```

[95]:
```
os.listdir(PATH)
```

[95]:
```
['.ipynb_checkpoints', 'archivo.zip', 'rideshare_kaggle.csv']
```

## 7 Cargamos el dataset

[96]:
```
Ask = pd.read_csv(PATH + '/rideshare_kaggle.csv')
```

[97]:
```
Ask.head()
```

[97]:

| | id | timestamp | hour | day | month | \ |
|---|---|---|---|---|---|---|
| 0 | 424553bb-7174-41ea-aeb4-fe06d4f4b9d7 | 1.544953e+09 | 9 | 16 | 12 | |
| 1 | 4bd23055-6827-41c6-b23b-3c491f24e74d | 1.543284e+09 | 2 | 27 | 11 | |
| 2 | 981a3613-77af-4620-a42a-0c0866077d1e | 1.543367e+09 | 1 | 28 | 11 | |
| 3 | c2d88af2-d278-4bfd-a8d0-29ca77cc5512 | 1.543554e+09 | 4 | 30 | 11 | |
| 4 | e0126e1f-8ca9-4f2e-82b3-50505a09db9a | 1.543463e+09 | 3 | 29 | 11 | |

| | datetime | timezone | source | destination | \ |
|---|---|---|---|---|---|
| 0 | 2018-12-16 09:30:07 | America/New_York | Haymarket Square | North Station | |
| 1 | 2018-11-27 02:00:23 | America/New_York | Haymarket Square | North Station | |
| 2 | 2018-11-28 01:00:22 | America/New_York | Haymarket Square | North Station | |
| 3 | 2018-11-30 04:53:02 | America/New_York | Haymarket Square | North Station | |
| 4 | 2018-11-29 03:49:20 | America/New_York | Haymarket Square | North Station | |

| | cab_type | … | precipIntensityMax | uvIndexTime | temperatureMin | \ |
|---|---|---|---|---|---|---|
| 0 | Lyft | … | 0.1276 | 1544979600 | 39.89 | |
| 1 | Lyft | … | 0.1300 | 1543251600 | 40.49 | |
| 2 | Lyft | … | 0.1064 | 1543338000 | 35.36 | |
| 3 | Lyft | … | 0.0000 | 1543507200 | 34.67 | |
| 4 | Lyft | … | 0.0001 | 1543420800 | 33.10 | |

| | temperatureMinTime | temperatureMax | temperatureMaxTime | \ |
|---|---|---|---|---|

```
0          1545012000           43.68          1544968800
1          1543233600           47.30          1543251600
2          1543377600           47.55          1543320000
3          1543550400           45.03          1543510800
4          1543402800           42.18          1543420800

   apparentTemperatureMin  apparentTemperatureMinTime  apparentTemperatureMax  \
0                   33.73                  1545012000                   38.07
1                   36.20                  1543291200                   43.92
2                   31.04                  1543377600                   44.12
3                   30.30                  1543550400                   38.53
4                   29.11                  1543392000                   35.75

  apparentTemperatureMaxTime
0                 1544958000
1                 1543251600
2                 1543320000
3                 1543510800
4                 1543420800

[5 rows x 57 columns]
```

# 8 2. Análisis Exploratorio de Datos (EDA)

Descripción de los datos

[98]:
```
Ask.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 693071 entries, 0 to 693070
Data columns (total 57 columns):
 #   Column                      Non-Null Count   Dtype
---  ------                      --------------   -----
 0   id                          693071 non-null  object
 1   timestamp                   693071 non-null  float64
 2   hour                        693071 non-null  int64
 3   day                         693071 non-null  int64
 4   month                       693071 non-null  int64
 5   datetime                    693071 non-null  object
 6   timezone                    693071 non-null  object
 7   source                      693071 non-null  object
 8   destination                 693071 non-null  object
 9   cab_type                    693071 non-null  object
 10  product_id                  693071 non-null  object
 11  name                        693071 non-null  object
 12  price                       637976 non-null  float64
 13  distance                    693071 non-null  float64
```

```
 14  surge_multiplier             693071 non-null  float64
 15  latitude                     693071 non-null  float64
 16  longitude                    693071 non-null  float64
 17  temperature                  693071 non-null  float64
 18  apparentTemperature          693071 non-null  float64
 19  short_summary                693071 non-null  object
 20  long_summary                 693071 non-null  object
 21  precipIntensity              693071 non-null  float64
 22  precipProbability            693071 non-null  float64
 23  humidity                     693071 non-null  float64
 24  windSpeed                    693071 non-null  float64
 25  windGust                     693071 non-null  float64
 26  windGustTime                 693071 non-null  int64
 27  visibility                   693071 non-null  float64
 28  temperatureHigh              693071 non-null  float64
 29  temperatureHighTime          693071 non-null  int64
 30  temperatureLow               693071 non-null  float64
 31  temperatureLowTime           693071 non-null  int64
 32  apparentTemperatureHigh      693071 non-null  float64
 33  apparentTemperatureHighTime  693071 non-null  int64
 34  apparentTemperatureLow       693071 non-null  float64
 35  apparentTemperatureLowTime   693071 non-null  int64
 36  icon                         693071 non-null  object
 37  dewPoint                     693071 non-null  float64
 38  pressure                     693071 non-null  float64
 39  windBearing                  693071 non-null  int64
 40  cloudCover                   693071 non-null  float64
 41  uvIndex                      693071 non-null  int64
 42  visibility.1                 693071 non-null  float64
 43  ozone                        693071 non-null  float64
 44  sunriseTime                  693071 non-null  int64
 45  sunsetTime                   693071 non-null  int64
 46  moonPhase                    693071 non-null  float64
 47  precipIntensityMax           693071 non-null  float64
 48  uvIndexTime                  693071 non-null  int64
 49  temperatureMin               693071 non-null  float64
 50  temperatureMinTime           693071 non-null  int64
 51  temperatureMax               693071 non-null  float64
 52  temperatureMaxTime           693071 non-null  int64
 53  apparentTemperatureMin       693071 non-null  float64
 54  apparentTemperatureMinTime   693071 non-null  int64
 55  apparentTemperatureMax       693071 non-null  float64
 56  apparentTemperatureMaxTime   693071 non-null  int64
dtypes: float64(29), int64(17), object(11)
memory usage: 301.4+ MB
```

[99]:
```
# Completamos los datos nulos de la columna price con la media
Ask.loc[:, 'price'].fillna(Ask['price'].mean(), inplace=True)
```

[100]:
```
Ask['price'] = Ask['price'].fillna(Ask['price'].mean())
```

[101]:
```
# Se depuran las columnas no requeridas
columns_to_drop = ['id', 'timestamp', 'datetime', 'timezone', 'product_id',␣
 ↪'latitude', 'longitude']
Ask.drop(columns=columns_to_drop, inplace=True)
```

[102]:
```
Ask.describe()
```

[102]:

| | hour | day | month | price | \ |
|---|---|---|---|---|---|
| count | 693071.000000 | 693071.000000 | 693071.000000 | 693071.000000 | |
| mean | 11.619137 | 17.794365 | 11.586684 | 16.545125 | |
| std | 6.948114 | 9.982286 | 0.492429 | 8.946069 | |
| min | 0.000000 | 1.000000 | 11.000000 | 2.500000 | |
| 25% | 6.000000 | 13.000000 | 11.000000 | 9.500000 | |
| 50% | 12.000000 | 17.000000 | 12.000000 | 16.000000 | |
| 75% | 18.000000 | 28.000000 | 12.000000 | 22.500000 | |
| max | 23.000000 | 30.000000 | 12.000000 | 97.500000 | |

| | distance | surge_multiplier | temperature | apparentTemperature | \ |
|---|---|---|---|---|---|
| count | 693071.000000 | 693071.000000 | 693071.000000 | 693071.000000 | |
| mean | 2.189430 | 1.013870 | 39.584388 | 35.881222 | |
| std | 1.138937 | 0.091641 | 6.726084 | 7.918707 | |
| min | 0.020000 | 1.000000 | 18.910000 | 12.130000 | |
| 25% | 1.280000 | 1.000000 | 36.450000 | 31.910000 | |
| 50% | 2.160000 | 1.000000 | 40.490000 | 35.900000 | |
| 75% | 2.920000 | 1.000000 | 43.580000 | 40.080000 | |
| max | 7.860000 | 3.000000 | 57.220000 | 57.220000 | |

| | precipIntensity | precipProbability | … | precipIntensityMax | \ |
|---|---|---|---|---|---|
| count | 693071.000000 | 693071.000000 | … | 693071.000000 | |
| mean | 0.008922 | 0.146076 | … | 0.037374 | |
| std | 0.026901 | 0.328938 | … | 0.055214 | |
| min | 0.000000 | 0.000000 | … | 0.000000 | |
| 25% | 0.000000 | 0.000000 | … | 0.000000 | |
| 50% | 0.000000 | 0.000000 | … | 0.000400 | |
| 75% | 0.000000 | 0.000000 | … | 0.091600 | |
| max | 0.144700 | 1.000000 | … | 0.145900 | |

| | uvIndexTime | temperatureMin | temperatureMinTime | temperatureMax | \ |
|---|---|---|---|---|---|
| count | 6.930710e+05 | 693071.000000 | 6.930710e+05 | 693071.000000 | |
| mean | 1.544044e+09 | 33.457774 | 1.544042e+09 | 45.261313 | |
| std | 6.912028e+05 | 6.467224 | 6.901954e+05 | 5.645046 | |
| min | 1.543162e+09 | 15.630000 | 1.543122e+09 | 33.510000 | |

```
25%    1.543421e+09       30.170000        1.543399e+09       42.570000
50%    1.543770e+09       34.240000        1.543727e+09       44.680000
75%    1.544807e+09       38.880000        1.544789e+09       46.910000
max    1.545152e+09       43.100000        1.545192e+09       57.870000

       temperatureMaxTime  apparentTemperatureMin  apparentTemperatureMinTime  \
count        6.930710e+05           693071.000000                6.930710e+05
mean         1.544047e+09               29.731002                1.544048e+09
std          6.901353e+05                7.110494                6.871862e+05
min          1.543154e+09               11.810000                1.543136e+09
25%          1.543439e+09               27.760000                1.543399e+09
50%          1.543788e+09               30.130000                1.543745e+09
75%          1.544814e+09               35.710000                1.544789e+09
max          1.545109e+09               40.050000                1.545134e+09

       apparentTemperatureMax  apparentTemperatureMaxTime
count           693071.000000                6.930710e+05
mean                41.997343                1.544048e+09
std                  6.936841                6.910777e+05
min                 28.950000                1.543187e+09
25%                 36.570000                1.543439e+09
50%                 40.950000                1.543788e+09
75%                 44.120000                1.544818e+09
max                 57.200000                1.545109e+09

[8 rows x 43 columns]
```

[103]:
```
Ask.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 693071 entries, 0 to 693070
Data columns (total 50 columns):
 #   Column                      Non-Null Count   Dtype
---  ------                      --------------   -----
 0   hour                        693071 non-null  int64
 1   day                         693071 non-null  int64
 2   month                       693071 non-null  int64
 3   source                      693071 non-null  object
 4   destination                 693071 non-null  object
 5   cab_type                    693071 non-null  object
 6   name                        693071 non-null  object
 7   price                       693071 non-null  float64
 8   distance                    693071 non-null  float64
 9   surge_multiplier            693071 non-null  float64
 10  temperature                 693071 non-null  float64
 11  apparentTemperature         693071 non-null  float64
 12  short_summary               693071 non-null  object
 13  long_summary                693071 non-null  object
```

```
 14  precipIntensity              693071 non-null  float64
 15  precipProbability            693071 non-null  float64
 16  humidity                     693071 non-null  float64
 17  windSpeed                    693071 non-null  float64
 18  windGust                     693071 non-null  float64
 19  windGustTime                 693071 non-null  int64
 20  visibility                   693071 non-null  float64
 21  temperatureHigh              693071 non-null  float64
 22  temperatureHighTime          693071 non-null  int64
 23  temperatureLow               693071 non-null  float64
 24  temperatureLowTime           693071 non-null  int64
 25  apparentTemperatureHigh      693071 non-null  float64
 26  apparentTemperatureHighTime  693071 non-null  int64
 27  apparentTemperatureLow       693071 non-null  float64
 28  apparentTemperatureLowTime   693071 non-null  int64
 29  icon                         693071 non-null  object
 30  dewPoint                     693071 non-null  float64
 31  pressure                     693071 non-null  float64
 32  windBearing                  693071 non-null  int64
 33  cloudCover                   693071 non-null  float64
 34  uvIndex                      693071 non-null  int64
 35  visibility.1                 693071 non-null  float64
 36  ozone                        693071 non-null  float64
 37  sunriseTime                  693071 non-null  int64
 38  sunsetTime                   693071 non-null  int64
 39  moonPhase                    693071 non-null  float64
 40  precipIntensityMax           693071 non-null  float64
 41  uvIndexTime                  693071 non-null  int64
 42  temperatureMin               693071 non-null  float64
 43  temperatureMinTime           693071 non-null  int64
 44  temperatureMax               693071 non-null  float64
 45  temperatureMaxTime           693071 non-null  int64
 46  apparentTemperatureMin       693071 non-null  float64
 47  apparentTemperatureMinTime   693071 non-null  int64
 48  apparentTemperatureMax       693071 non-null  float64
 49  apparentTemperatureMaxTime   693071 non-null  int64
dtypes: float64(26), int64(17), object(7)
memory usage: 264.4+ MB
```

[104]:
```
from sklearn.preprocessing import LabelEncoder

# Se transforman los datos numéricos
columnas_no_numericas=['source', 'destination', 'cab_type', 'name',␣
 ↪'short_summary', 'long_summary', 'icon']

encoder = LabelEncoder()
for columna in columnas_no_numericas:
```

```
    Ask[columna] = encoder.fit_transform(Ask[columna])
```

[105]:
```
# Se aplica la correlación
correlacion = Ask.corr()
correlacion
```

[105]:

| | hour | day | month | source | \ |
|---|---|---|---|---|---|
| hour | 1.000000 | 0.066090 | -0.079379 | 0.002685 | |
| day | 0.066090 | 1.000000 | -0.861306 | -0.000760 | |
| month | -0.079379 | -0.861306 | 1.000000 | 0.000174 | |
| source | 0.002685 | -0.000760 | 0.000174 | 1.000000 | |
| destination | -0.004190 | -0.001813 | 0.000670 | -0.007584 | |
| cab_type | -0.001257 | 0.001868 | -0.000776 | 0.000912 | |
| name | 0.000158 | 0.000993 | -0.000106 | 0.000335 | |
| price | 0.000538 | -0.000693 | 0.000876 | -0.024336 | |
| distance | 0.002280 | -0.000627 | 0.002298 | -0.091369 | |
| surge_multiplier | -0.000077 | 0.001389 | -0.000628 | -0.032521 | |
| temperature | 0.218769 | -0.084116 | -0.081774 | -0.000388 | |
| apparentTemperature | 0.198190 | -0.223545 | 0.024983 | -0.000070 | |
| short_summary | 0.120234 | -0.011329 | -0.024080 | -0.000642 | |
| long_summary | -0.045528 | 0.306418 | -0.146440 | -0.000972 | |
| precipIntensity | -0.233349 | 0.091664 | -0.154531 | -0.003992 | |
| precipProbability | -0.129725 | 0.030903 | -0.098031 | -0.002768 | |
| humidity | -0.275762 | -0.093340 | -0.022256 | -0.001253 | |
| windSpeed | 0.072778 | 0.440502 | -0.334803 | -0.001113 | |
| windGust | 0.077308 | 0.402637 | -0.270494 | -0.000304 | |
| windGustTime | -0.028047 | -0.336417 | 0.767265 | -0.000647 | |
| visibility | 0.174129 | 0.104992 | 0.011493 | 0.000175 | |
| temperatureHigh | 0.005172 | -0.283561 | 0.068313 | 0.000102 | |
| temperatureHighTime | -0.027838 | -0.343118 | 0.772053 | -0.000541 | |
| temperatureLow | 0.015326 | -0.088320 | -0.138732 | -0.003518 | |
| temperatureLowTime | -0.026797 | -0.333558 | 0.765489 | -0.000459 | |
| apparentTemperatureHigh | 0.005513 | -0.398183 | 0.152651 | -0.000154 | |
| apparentTemperatureHighTime | -0.027997 | -0.343291 | 0.772163 | -0.000533 | |
| apparentTemperatureLow | 0.016005 | -0.192984 | -0.086998 | -0.002534 | |
| apparentTemperatureLowTime | -0.026970 | -0.335504 | 0.766822 | -0.000462 | |
| icon | -0.041443 | -0.019885 | -0.073223 | -0.003292 | |
| dewPoint | 0.004839 | -0.101582 | -0.074794 | -0.001025 | |
| pressure | -0.067366 | -0.444635 | 0.589609 | -0.001169 | |
| windBearing | 0.036089 | 0.092167 | -0.135103 | 0.003904 | |
| cloudCover | 0.026525 | -0.041190 | -0.016800 | -0.001134 | |
| uvIndex | 0.330209 | -0.011885 | -0.012321 | -0.001664 | |
| visibility.1 | 0.174129 | 0.104992 | 0.011493 | 0.000175 | |
| ozone | 0.034164 | 0.377057 | -0.149333 | 0.001013 | |
| sunriseTime | -0.027537 | -0.341068 | 0.770684 | -0.000519 | |
| sunsetTime | -0.027535 | -0.341012 | 0.770646 | -0.000519 | |
| moonPhase | 0.020483 | -0.007260 | -0.452469 | 0.002514 | |

```
precipIntensityMax           0.019940  0.145568 -0.213054  0.000411
uvIndexTime                 -0.027518 -0.341198  0.770767 -0.000515
temperatureMin               0.015513  0.038076 -0.191803  0.000545
temperatureMinTime          -0.027027 -0.333990  0.765255 -0.000349
temperatureMax               0.002958 -0.304870  0.105326  0.000599
temperatureMaxTime          -0.027990 -0.344532  0.772989 -0.000595
apparentTemperatureMin       0.015820 -0.019940 -0.191153 -0.000112
apparentTemperatureMinTime  -0.027704 -0.335043  0.766031 -0.000417
apparentTemperatureMax       0.002897 -0.444445  0.215076  0.000514
apparentTemperatureMaxTime  -0.027785 -0.344411  0.772896 -0.000573

                            destination  cab_type      name     price  \
hour                          -0.004190 -0.001257  0.000158  0.000538
day                           -0.001813  0.001868  0.000993 -0.000693
month                          0.000670 -0.000776 -0.000106  0.000876
source                        -0.007584  0.000912  0.000335 -0.024336
destination                    1.000000  0.000835  0.000307 -0.044439
cab_type                       0.000835  1.000000  0.366068 -0.080464
name                           0.000307  0.366068  1.000000 -0.572357
price                         -0.044439 -0.080464 -0.572357  1.000000
distance                      -0.147394  0.001924  0.000705  0.330038
surge_multiplier              -0.001027 -0.169522 -0.081516  0.240220
temperature                   -0.000758 -0.001592 -0.001657 -0.000081
apparentTemperature            0.000136 -0.002651 -0.002056 -0.000185
short_summary                 -0.003483 -0.000458 -0.001197  0.001194
long_summary                  -0.000211  0.001620 -0.000202  0.000380
precipIntensity               -0.002655  0.001908 -0.000888  0.000159
precipProbability             -0.003744  0.002746 -0.000198 -0.000233
humidity                       0.001856  0.002476  0.000231 -0.001187
windSpeed                     -0.003163  0.003018  0.000615  0.000950
windGust                      -0.001865  0.002510  0.000340  0.001168
windGustTime                  -0.001227  0.000865  0.001012  0.000817
visibility                     0.003333 -0.002026 -0.000110  0.001435
temperatureHigh                0.001286 -0.002967 -0.002132 -0.000495
temperatureHighTime           -0.001262  0.000889  0.001130  0.000728
temperatureLow                 0.000279 -0.003674 -0.003334  0.000175
temperatureLowTime            -0.001229  0.000946  0.001167  0.000723
apparentTemperatureHigh        0.002565 -0.003400 -0.002440 -0.000250
apparentTemperatureHighTime   -0.001240  0.000881  0.001136  0.000711
apparentTemperatureLow         0.001954 -0.003611 -0.003701  0.000797
apparentTemperatureLowTime    -0.001182  0.000933  0.001160  0.000731
icon                          -0.004879  0.002268 -0.000893  0.002015
dewPoint                       0.000572  0.000124 -0.001088 -0.000590
pressure                       0.001868 -0.000486 -0.000568  0.000831
windBearing                    0.003161 -0.001235  0.001309 -0.001326
cloudCover                    -0.001964  0.001140 -0.000843  0.000847
uvIndex                       -0.002622 -0.000764  0.000094 -0.000407
```

```
visibility.1                  0.003333 -0.002026 -0.000110  0.001435
ozone                        -0.003325  0.000388  0.001266  0.000442
sunriseTime                  -0.001223  0.000898  0.001123  0.000729
sunsetTime                   -0.001224  0.000898  0.001123  0.000729
moonPhase                     0.000597 -0.001490 -0.000599 -0.001537
precipIntensityMax            0.000536 -0.000241 -0.001628  0.001069
uvIndexTime                  -0.001222  0.000905  0.001123  0.000732
temperatureMin               -0.000603 -0.001249 -0.001127 -0.000411
temperatureMinTime           -0.001170  0.000940  0.001119  0.000738
temperatureMax                0.000851 -0.002849 -0.002061 -0.000527
temperatureMaxTime           -0.001226  0.000874  0.001121  0.000735
apparentTemperatureMin        0.001010 -0.002211 -0.001698 -0.000556
apparentTemperatureMinTime   -0.001163  0.000902  0.001066  0.000781
apparentTemperatureMax        0.002080 -0.003340 -0.002416 -0.000278
apparentTemperatureMaxTime   -0.001204  0.000857  0.001124  0.000716

                            distance  surge_multiplier  …  \
hour                        0.002280         -0.000077  …
day                        -0.000627          0.001389  …
month                       0.002298         -0.000628  …
source                     -0.091369         -0.032521  …
destination                -0.147394         -0.001027  …
cab_type                    0.001924         -0.169522  …
name                        0.000705         -0.081516  …
price                       0.330038          0.240220  …
distance                    1.000000          0.024769  …
surge_multiplier            0.024769          1.000000  …
temperature                -0.002884         -0.001572  …
apparentTemperature        -0.003116         -0.002532  …
short_summary               0.000636         -0.002215  …
long_summary                0.003215         -0.003738  …
precipIntensity            -0.000256         -0.001530  …
precipProbability           0.000371         -0.002769  …
humidity                   -0.003901         -0.001676  …
windSpeed                   0.002277          0.002511  …
windGust                    0.001364          0.002093  …
windGustTime                0.003618          0.000439  …
visibility                  0.004092          0.001714  …
temperatureHigh            -0.004154         -0.001897  …
temperatureHighTime         0.003604          0.000520  …
temperatureLow             -0.002146         -0.002601  …
temperatureLowTime          0.003556          0.000573  …
apparentTemperatureHigh    -0.004098         -0.002041  …
apparentTemperatureHighTime 0.003585          0.000520  …
apparentTemperatureLow     -0.001501         -0.002994  …
apparentTemperatureLowTime  0.003583          0.000551  …
icon                        0.001684          0.000330  …
```

```
dewPoint                   -0.004078         -0.001849  …
pressure                    0.003490         -0.003003  …
windBearing                -0.002146          0.002167  …
cloudCover                 -0.000905         -0.002103  …
uvIndex                     0.002318         -0.002422  …
visibility.1                0.004092          0.001714  …
ozone                       0.001467          0.003324  …
sunriseTime                 0.003556          0.000544  …
sunsetTime                  0.003556          0.000544  …
moonPhase                  -0.004594         -0.000150  …
precipIntensityMax          0.000648         -0.000101  …
uvIndexTime                 0.003562          0.000544  …
temperatureMin             -0.003396         -0.000724  …
temperatureMinTime          0.003428          0.000554  …
temperatureMax             -0.004191         -0.001951  …
temperatureMaxTime          0.003580          0.000522  …
apparentTemperatureMin     -0.003937         -0.001304  …
apparentTemperatureMinTime  0.003526          0.000464  …
apparentTemperatureMax     -0.004238         -0.002154  …
apparentTemperatureMaxTime  0.003584          0.000519  …

                           precipIntensityMax  uvIndexTime  temperatureMin  \
hour                                 0.019940    -0.027518        0.015513
day                                  0.145568    -0.341198        0.038076
month                               -0.213054     0.770767       -0.191803
source                               0.000411    -0.000515        0.000545
destination                          0.000536    -0.001222       -0.000603
cab_type                            -0.000241     0.000905       -0.001249
name                                -0.001628     0.001123       -0.001127
price                                0.001069     0.000732       -0.000411
distance                             0.000648     0.003562       -0.003396
surge_multiplier                    -0.000101     0.000544       -0.000724
temperature                          0.333081    -0.254467        0.789261
apparentTemperature                  0.257072    -0.231309        0.697441
short_summary                        0.145570    -0.058214        0.208596
long_summary                         0.343737     0.107158       -0.046292
precipIntensity                      0.498788    -0.180224        0.236578
precipProbability                    0.583335    -0.144637        0.274337
humidity                             0.520122    -0.160387        0.461731
windSpeed                            0.212282    -0.067981        0.173915
windGust                             0.100243     0.005006        0.081142
windGustTime                        -0.201832     0.999362       -0.300956
visibility                          -0.545052     0.152129       -0.352869
temperatureHigh                      0.192505    -0.228204        0.794548
temperatureHighTime                 -0.214441     0.999909       -0.309750
temperatureLow                       0.336297    -0.369013        0.473264
temperatureLowTime                  -0.210034     0.999805       -0.305101
```

| | | | |
|---|---|---|---|
| apparentTemperatureHigh | 0.173295 | -0.215608 | 0.665066 |
| apparentTemperatureHighTime | -0.216641 | 0.999895 | -0.312110 |
| apparentTemperatureLow | 0.254630 | -0.403192 | 0.284451 |
| apparentTemperatureLowTime | -0.207548 | 0.999831 | -0.304969 |
| icon | 0.228491 | -0.161686 | 0.191049 |
| dewPoint | 0.505994 | -0.265557 | 0.797175 |
| pressure | -0.243383 | 0.533110 | -0.454733 |
| windBearing | -0.555013 | -0.133758 | -0.251004 |
| cloudCover | 0.396136 | -0.080241 | 0.284974 |
| uvIndex | -0.040818 | -0.023841 | -0.022146 |
| visibility.1 | -0.545052 | 0.152129 | -0.352869 |
| ozone | -0.223173 | 0.193620 | -0.183328 |
| sunriseTime | -0.210650 | 0.999997 | -0.307699 |
| sunsetTime | -0.210612 | 0.999997 | -0.307682 |
| moonPhase | 0.055632 | -0.844775 | 0.401172 |
| precipIntensityMax | 1.000000 | -0.209327 | 0.446451 |
| uvIndexTime | -0.209327 | 1.000000 | -0.307080 |
| temperatureMin | 0.446451 | -0.307080 | 1.000000 |
| temperatureMinTime | -0.209823 | 0.999016 | -0.294607 |
| temperatureMax | 0.180298 | -0.185383 | 0.782945 |
| temperatureMaxTime | -0.213461 | 0.999780 | -0.306577 |
| apparentTemperatureMin | 0.409633 | -0.379697 | 0.953854 |
| apparentTemperatureMinTime | -0.188085 | 0.999010 | -0.278444 |
| apparentTemperatureMax | 0.157479 | -0.156584 | 0.648381 |
| apparentTemperatureMaxTime | -0.215784 | 0.999776 | -0.308665 |

| | temperatureMinTime | temperatureMax | \ |
|---|---|---|---|
| hour | -0.027027 | 0.002958 | |
| day | -0.333990 | -0.304870 | |
| month | 0.765255 | 0.105326 | |
| source | -0.000349 | 0.000599 | |
| destination | -0.001170 | 0.000851 | |
| cab_type | 0.000940 | -0.002849 | |
| name | 0.001119 | -0.002061 | |
| price | 0.000738 | -0.000527 | |
| distance | 0.003428 | -0.004191 | |
| surge_multiplier | 0.000554 | -0.001951 | |
| temperature | -0.242698 | 0.780546 | |
| apparentTemperature | -0.224744 | 0.792625 | |
| short_summary | -0.061117 | 0.098440 | |
| long_summary | 0.093727 | -0.257097 | |
| precipIntensity | -0.182233 | 0.114020 | |
| precipProbability | -0.147988 | 0.142092 | |
| humidity | -0.167838 | 0.394811 | |
| windSpeed | -0.051350 | -0.120832 | |
| windGust | 0.023669 | -0.144780 | |
| windGustTime | 0.998324 | -0.179722 | |

```
visibility                      0.152441       -0.297159
temperatureHigh                -0.214214        0.986001
temperatureHighTime             0.998486       -0.188793
temperatureLow                 -0.383542        0.567568
temperatureLowTime              0.999147       -0.188336
apparentTemperatureHigh        -0.206671        0.926454
apparentTemperatureHighTime     0.998464       -0.189612
apparentTemperatureLow         -0.416424        0.492281
apparentTemperatureLowTime      0.999054       -0.187624
icon                           -0.162437        0.096632
dewPoint                       -0.261520        0.751152
pressure                        0.519971       -0.164320
windBearing                    -0.127096       -0.166590
cloudCover                     -0.087445        0.114756
uvIndex                        -0.024780       -0.007029
visibility.1                    0.152441       -0.297159
ozone                           0.199331       -0.444013
sunriseTime                     0.999000       -0.185870
sunsetTime                      0.999001       -0.185891
moonPhase                      -0.841619        0.372091
precipIntensityMax             -0.209823        0.180298
uvIndexTime                     0.999016       -0.185383
temperatureMin                 -0.294607        0.782945
temperatureMinTime              1.000000       -0.168071
temperatureMax                 -0.168071        1.000000
temperatureMaxTime              0.998165       -0.185512
apparentTemperatureMin         -0.372710        0.838358
apparentTemperatureMinTime      0.999376       -0.162140
apparentTemperatureMax         -0.143231        0.956358
apparentTemperatureMaxTime      0.998136       -0.186660

                           temperatureMaxTime  apparentTemperatureMin  \
hour                               -0.027990                0.015820
day                                -0.344532               -0.019940
month                               0.772989               -0.191153
source                             -0.000595               -0.000112
destination                        -0.001226                0.001010
cab_type                            0.000874               -0.002211
name                                0.001121               -0.001698
price                               0.000735               -0.000556
distance                            0.003580               -0.003937
surge_multiplier                    0.000522               -0.001304
temperature                        -0.253881                0.817521
apparentTemperature                -0.227712                0.786577
short_summary                      -0.057815                0.175481
long_summary                        0.111153               -0.032584
precipIntensity                    -0.182098                0.224691
```

| | | |
|---|---|---|
| precipProbability | -0.146557 | 0.263250 |
| humidity | -0.158278 | 0.512270 |
| windSpeed | -0.078361 | -0.020056 |
| windGust | -0.005769 | -0.116743 |
| windGustTime | 0.999302 | -0.371504 |
| visibility | 0.154674 | -0.344016 |
| temperatureHigh | -0.225931 | 0.877466 |
| temperatureHighTime | 0.999885 | -0.381387 |
| temperatureLow | -0.360320 | 0.636362 |
| temperatureLowTime | 0.999523 | -0.380357 |
| apparentTemperatureHigh | -0.211503 | 0.791976 |
| apparentTemperatureHighTime | 0.999870 | -0.383148 |
| apparentTemperatureLow | -0.394981 | 0.479460 |
| apparentTemperatureLowTime | 0.999618 | -0.378749 |
| icon | -0.162200 | 0.167114 |
| dewPoint | -0.263786 | 0.845406 |
| pressure | 0.539115 | -0.342568 |
| windBearing | -0.136100 | -0.268326 |
| cloudCover | -0.080245 | 0.249958 |
| uvIndex | -0.024287 | -0.019354 |
| visibility.1 | 0.154674 | -0.344016 |
| ozone | 0.189874 | -0.364767 |
| sunriseTime | 0.999793 | -0.380239 |
| sunsetTime | 0.999792 | -0.380239 |
| moonPhase | -0.845208 | 0.445199 |
| precipIntensityMax | -0.213461 | 0.409633 |
| uvIndexTime | 0.999780 | -0.379697 |
| temperatureMin | -0.306577 | 0.953854 |
| temperatureMinTime | 0.998165 | -0.372710 |
| temperatureMax | -0.185512 | 0.838358 |
| temperatureMaxTime | 1.000000 | -0.375731 |
| apparentTemperatureMin | -0.375731 | 1.000000 |
| apparentTemperatureMinTime | 0.998467 | -0.355199 |
| apparentTemperatureMax | -0.155643 | 0.741950 |
| apparentTemperatureMaxTime | 0.999959 | -0.377402 |

| | apparentTemperatureMinTime \ |
|---|---|
| hour | -0.027704 |
| day | -0.335043 |
| month | 0.766031 |
| source | -0.000417 |
| destination | -0.001163 |
| cab_type | 0.000902 |
| name | 0.001066 |
| price | 0.000781 |
| distance | 0.003526 |
| surge_multiplier | 0.000464 |

| | |
|---|---|
| temperature | -0.231513 |
| apparentTemperature | -0.212453 |
| short_summary | -0.054880 |
| long_summary | 0.115909 |
| precipIntensity | -0.170430 |
| precipProbability | -0.135967 |
| humidity | -0.152785 |
| windSpeed | -0.056391 |
| windGust | 0.014164 |
| windGustTime | 0.998958 |
| visibility | 0.142179 |
| temperatureHigh | -0.205708 |
| temperatureHighTime | 0.998537 |
| temperatureLow | -0.367590 |
| temperatureLowTime | 0.999023 |
| apparentTemperatureHigh | -0.197581 |
| apparentTemperatureHighTime | 0.998428 |
| apparentTemperatureLow | -0.404296 |
| apparentTemperatureLowTime | 0.999083 |
| icon | -0.158868 |
| dewPoint | -0.245568 |
| pressure | 0.526319 |
| windBearing | -0.148829 |
| cloudCover | -0.077754 |
| uvIndex | -0.026580 |
| visibility.1 | 0.142179 |
| ozone | 0.188392 |
| sunriseTime | 0.998971 |
| sunsetTime | 0.998972 |
| moonPhase | -0.845925 |
| precipIntensityMax | -0.188085 |
| uvIndexTime | 0.999010 |
| temperatureMin | -0.278444 |
| temperatureMinTime | 0.999376 |
| temperatureMax | -0.162140 |
| temperatureMaxTime | 0.998467 |
| apparentTemperatureMin | -0.355199 |
| apparentTemperatureMinTime | 1.000000 |
| apparentTemperatureMax | -0.137621 |
| apparentTemperatureMaxTime | 0.998345 |

| | apparentTemperatureMax \ |
|---|---|
| hour | 0.002897 |
| day | -0.444445 |
| month | 0.215076 |
| source | 0.000514 |
| destination | 0.002080 |

| | |
|---|---|
| cab_type | -0.003340 |
| name | -0.002416 |
| price | -0.000278 |
| distance | -0.004238 |
| surge_multiplier | -0.002154 |
| temperature | 0.703208 |
| apparentTemperature | 0.782396 |
| short_summary | 0.065379 |
| long_summary | -0.284568 |
| precipIntensity | 0.108896 |
| precipProbability | 0.136422 |
| humidity | 0.410480 |
| windSpeed | -0.304547 |
| windGust | -0.316410 |
| windGustTime | -0.151696 |
| visibility | -0.297751 |
| temperatureHigh | 0.951057 |
| temperatureHighTime | -0.159726 |
| temperatureLow | 0.579424 |
| temperatureLowTime | -0.161419 |
| apparentTemperatureHigh | 0.974289 |
| apparentTemperatureHighTime | -0.160064 |
| apparentTemperatureLow | 0.558193 |
| apparentTemperatureLowTime | -0.159448 |
| icon | 0.067454 |
| dewPoint | 0.705068 |
| pressure | 0.015840 |
| windBearing | -0.209258 |
| cloudCover | 0.090372 |
| uvIndex | 0.006063 |
| visibility.1 | -0.297751 |
| ozone | -0.591933 |
| sunriseTime | -0.156994 |
| sunsetTime | -0.157028 |
| moonPhase | 0.344305 |
| precipIntensityMax | 0.157479 |
| uvIndexTime | -0.156584 |
| temperatureMin | 0.648381 |
| temperatureMinTime | -0.143231 |
| temperatureMax | 0.956358 |
| temperatureMaxTime | -0.155643 |
| apparentTemperatureMin | 0.741950 |
| apparentTemperatureMinTime | -0.137621 |
| apparentTemperatureMax | 1.000000 |
| apparentTemperatureMaxTime | -0.156426 |

apparentTemperatureMaxTime

| | |
|---|---|
| hour | -0.027785 |
| day | -0.344411 |
| month | 0.772896 |
| source | -0.000573 |
| destination | -0.001204 |
| cab_type | 0.000857 |
| name | 0.001124 |
| price | 0.000716 |
| distance | 0.003584 |
| surge_multiplier | 0.000519 |
| temperature | -0.255288 |
| apparentTemperature | -0.228674 |
| short_summary | -0.058945 |
| long_summary | 0.108549 |
| precipIntensity | -0.183115 |
| precipProbability | -0.147471 |
| humidity | -0.158422 |
| windSpeed | -0.079521 |
| windGust | -0.007036 |
| windGustTime | 0.999174 |
| visibility | 0.155300 |
| temperatureHigh | -0.226911 |
| temperatureHighTime | 0.999885 |
| temperatureLow | -0.361987 |
| temperatureLowTime | 0.999526 |
| apparentTemperatureHigh | -0.212064 |
| apparentTemperatureHighTime | 0.999901 |
| apparentTemperatureLow | -0.396409 |
| apparentTemperatureLowTime | 0.999632 |
| icon | -0.163354 |
| dewPoint | -0.264869 |
| pressure | 0.539298 |
| windBearing | -0.133472 |
| cloudCover | -0.081034 |
| uvIndex | -0.023411 |
| visibility.1 | 0.155300 |
| ozone | 0.189240 |
| sunriseTime | 0.999794 |
| sunsetTime | 0.999793 |
| moonPhase | -0.844708 |
| precipIntensityMax | -0.215784 |
| uvIndexTime | 0.999776 |
| temperatureMin | -0.308665 |
| temperatureMinTime | 0.998136 |
| temperatureMax | -0.186660 |
| temperatureMaxTime | 0.999959 |
| apparentTemperatureMin | -0.377402 |

```
apparentTemperatureMinTime                  0.998345
apparentTemperatureMax                     -0.156426
apparentTemperatureMaxTime                  1.000000

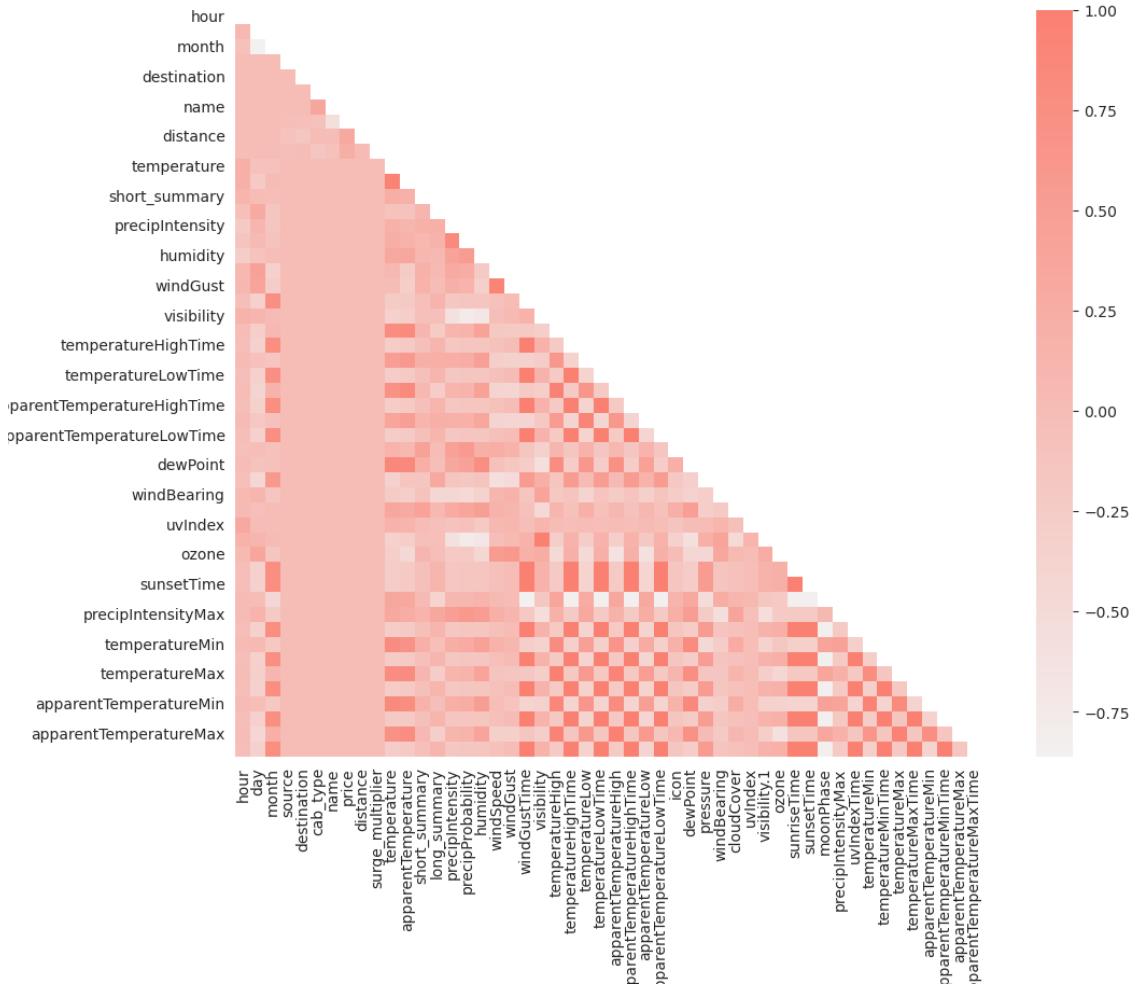
[50 rows x 50 columns]
```

No tenemos datos faltantes. No se rellenan datos. Se determina que los datos que se encuentran en diferentes escalas, por lo tanto es necesario escalar los datos

## 9 3. Visualización de datos

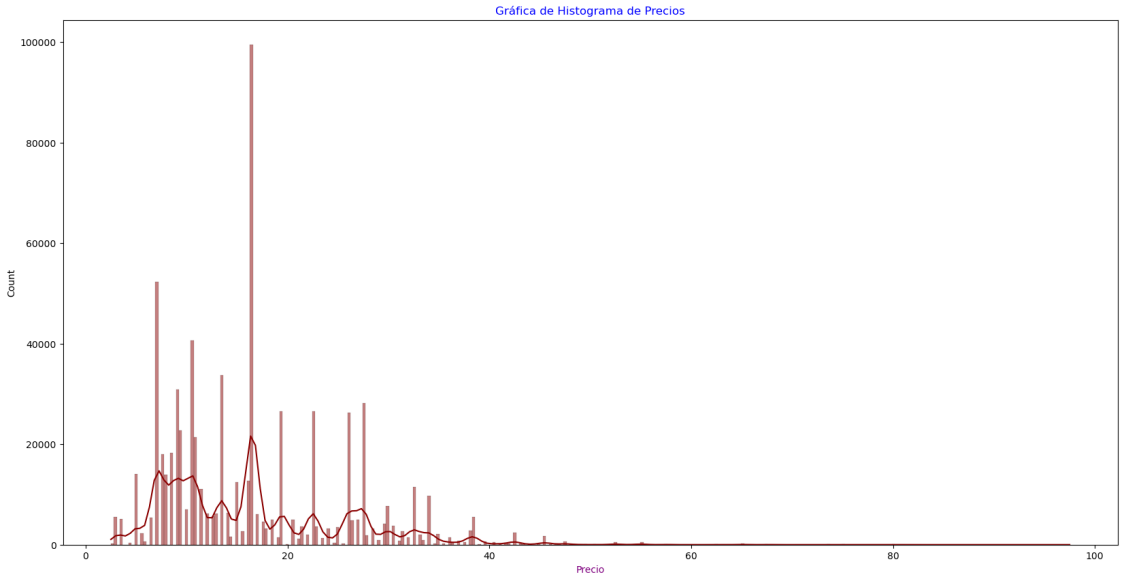
[106]:
```
import seaborn as sns
```

[107]:
```
# Se visualiza la correlación
cores = sns.color_palette('light:salmon', as_cmap=True)
mask = np.zeros_like(correlacion)
mask[np.triu_indices_from(mask)] = True
with sns.axes_style("white"):
    f, ax = plt.subplots(figsize=(13,9))
    ax = sns.heatmap(correlacion, cmap=cores, mask=mask, square=True,)
```



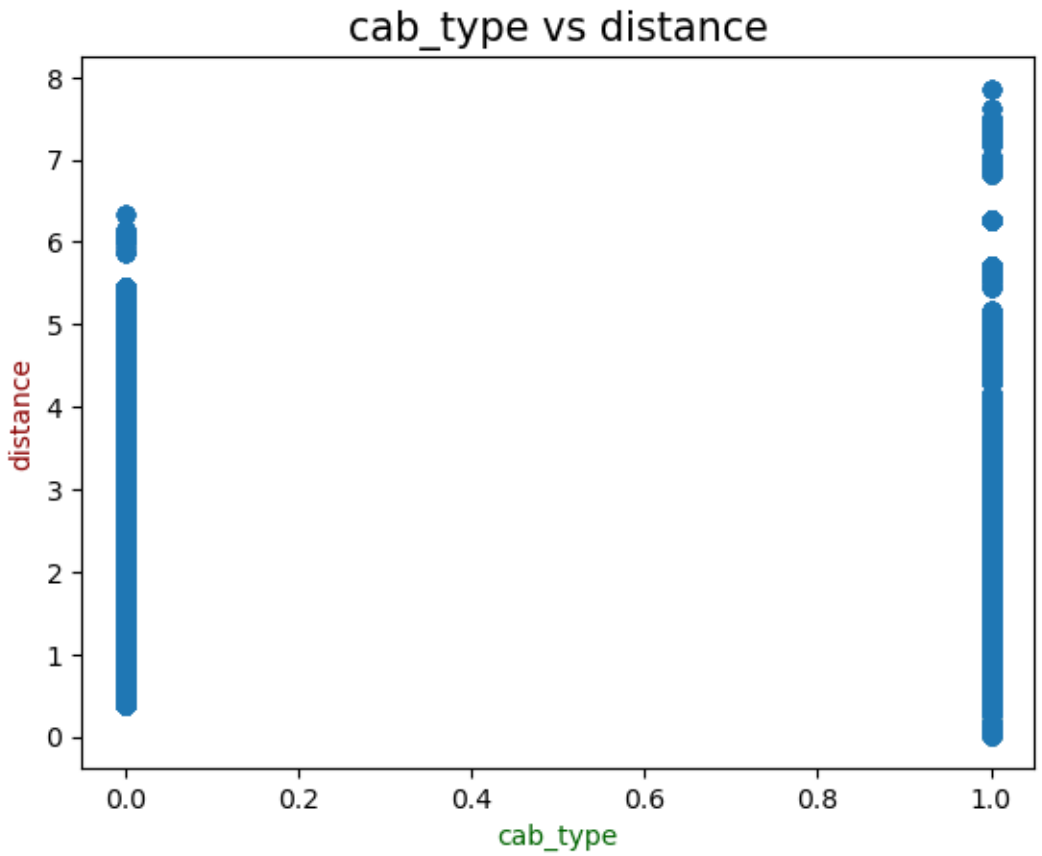

[108]:
```
ax = sns.histplot(data=Ask, x='price', kde=True, color='darkred')
ax.figure.set_size_inches(20,10)
ax.set_title('Gráfica de Histograma de Precios', color='blue')
ax.set_xlabel('Precio', color='purple')
```

[108]: Text(0.5, 0, 'Precio')

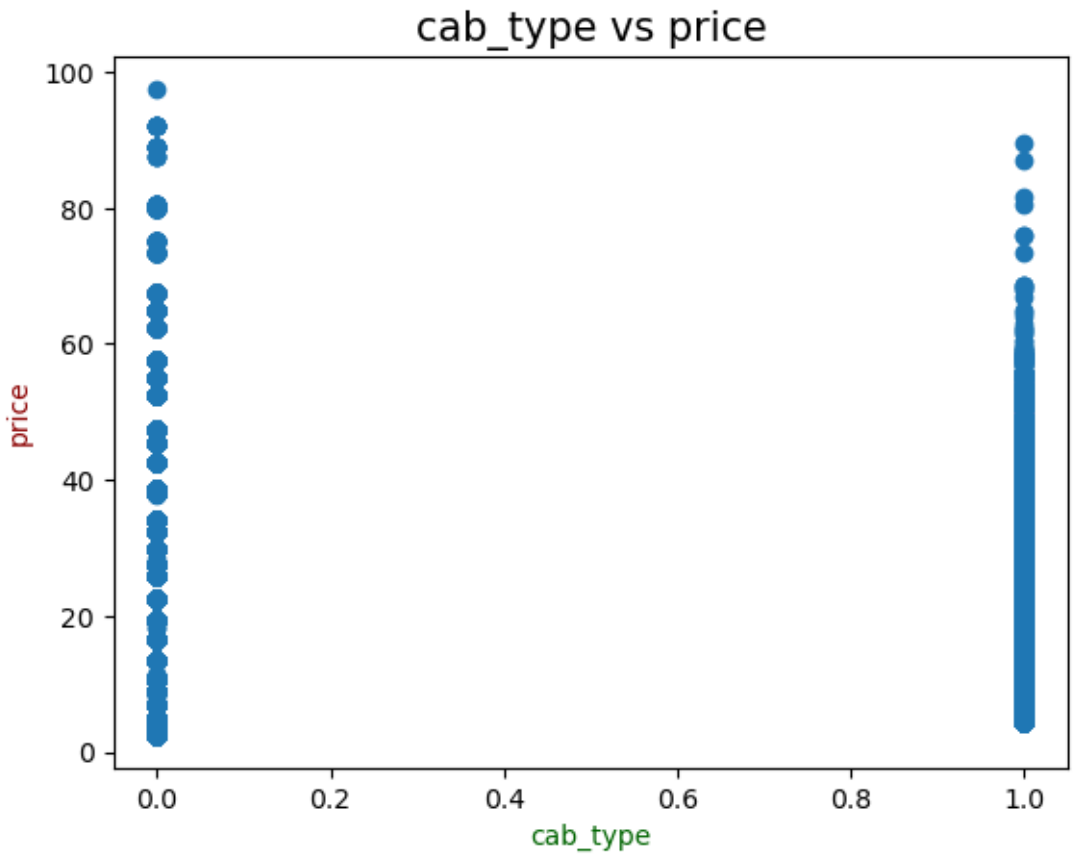



[109]:
```
def grafica(data, colx, coly, color='brown'):
    plt.plot(data[colx], data[coly], 'o')
    plt.xlabel(colx, size=10, color='darkgreen')
    plt.ylabel(coly, size=10, color='darkred')
    plt.title(colx +' vs ' + coly, size=15)
    # plt.title('Tipo de Servicio contra la Distancia', size = 15,␣
 ↪color='orange', fontweight='bold')
    plt.show()
```

[110]:
```
grafica(Ask, 'cab_type', 'distance', )
```

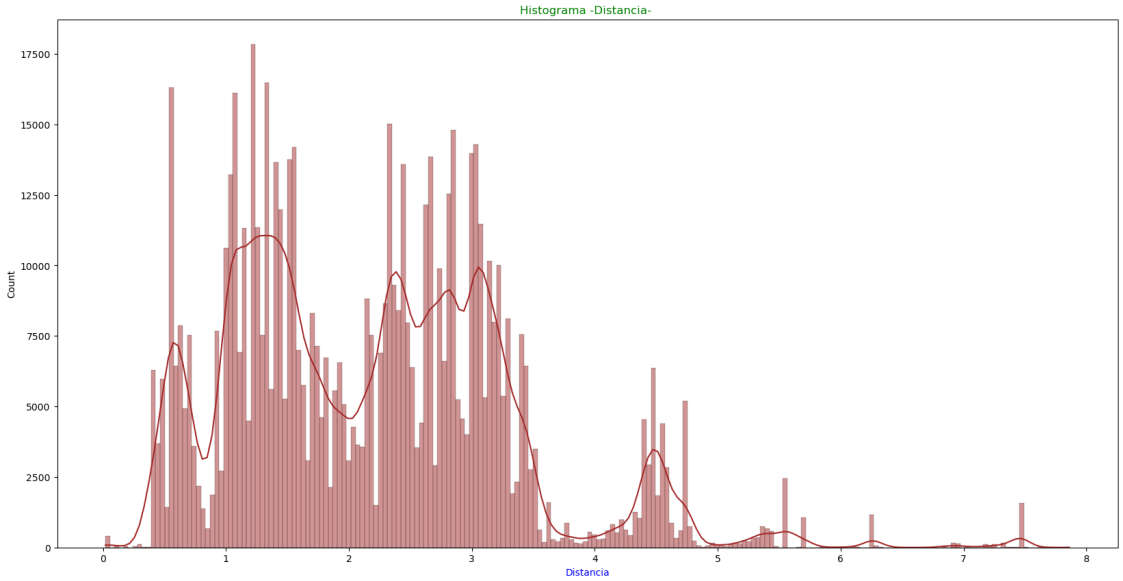



[111]:
```
grafica(Ask, 'cab_type', 'price')
```




[112]:
```
ax = sns.histplot(data=Ask, x='distance', kde=True, color='brown')
ax.figure.set_size_inches(20,10)
ax.set_title('Histograma -Distancia- ', color='green')
ax.set_xlabel('Distancia', color='blue')
```

[112]: Text(0.5, 0, 'Distancia')



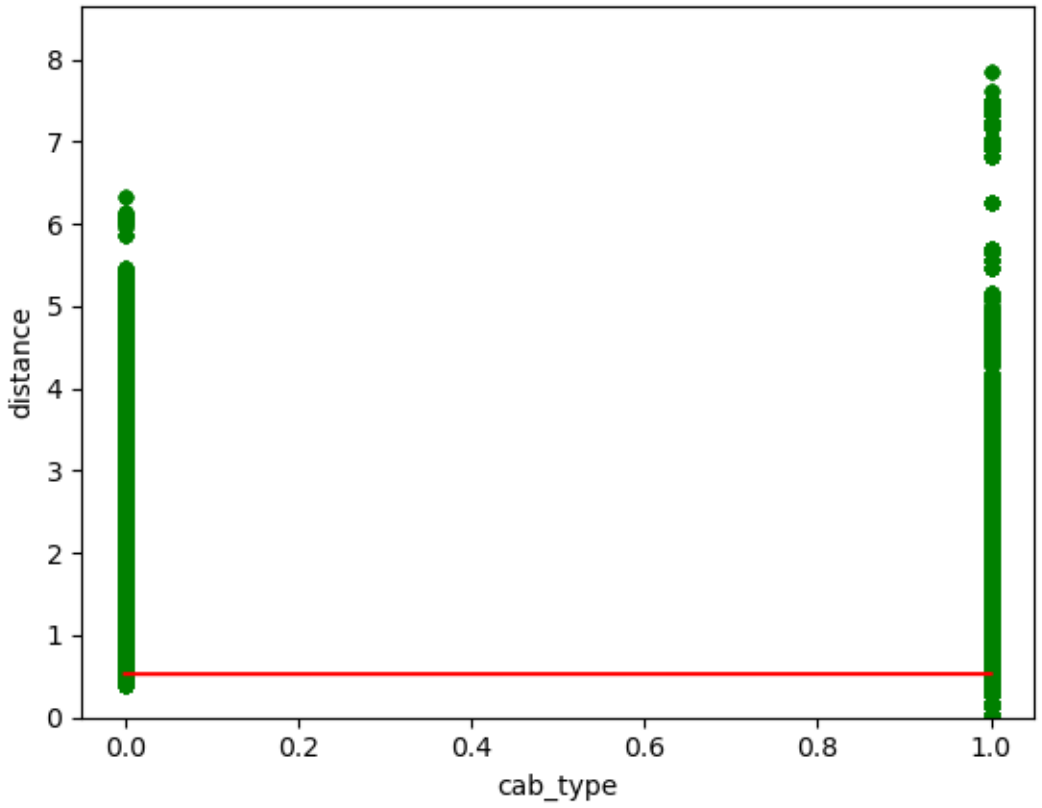

Los datos se concentran en la primera mitad de las variables independientes

[113]:
```
w = 0.005
b = 0.1
```

[127]:
```
# Puntos de la recta
x = np.linspace(0,Ask['cab_type'].max(),100)
y = 1 / (1 + np.exp(-(w * x + b)))
# Gráfica de la recta
Ask.plot.scatter(x='cab_type', y='distance', c='green')
plt.plot(x,y,'-r')
plt.ylim(0,Ask['distance'].max()*1.1)

plt.show
```

[127]:
```
<function matplotlib.pyplot.show(close=None, block=None)>
```




## 10 4. Crear la matriz de atributos $X$ y el vector de etiquetas $y$

[115]:
```
y = Ask['cab_type']
X = Ask.iloc[:, 1:]
```

[116]:
```
X.head()
```

[116]:

| | day | month | source | destination | cab_type | name | price | distance | \ |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 16 | 12 | 5 | 7 | 0 | 7 | 5.0 | 0.44 | |
| 1 | 27 | 11 | 5 | 7 | 0 | 2 | 11.0 | 0.44 | |
| 2 | 28 | 11 | 5 | 7 | 0 | 5 | 7.0 | 0.44 | |
| 3 | 30 | 11 | 5 | 7 | 0 | 4 | 26.0 | 0.44 | |
| 4 | 29 | 11 | 5 | 7 | 0 | 6 | 9.0 | 0.44 | |

| | surge_multiplier | temperature | … | precipIntensityMax | uvIndexTime | \ |
|---|---|---|---|---|---|---|
| 0 | 1.0 | 42.34 | … | 0.1276 | 1544979600 | |
| 1 | 1.0 | 43.58 | … | 0.1300 | 1543251600 | |
| 2 | 1.0 | 38.33 | … | 0.1064 | 1543338000 | |
| 3 | 1.0 | 34.38 | … | 0.0000 | 1543507200 | |
| 4 | 1.0 | 37.44 | … | 0.0001 | 1543420800 | |

```
   temperatureMin  temperatureMinTime  temperatureMax  temperatureMaxTime  \
0           39.89          1545012000           43.68          1544968800
1           40.49          1543233600           47.30          1543251600
2           35.36          1543377600           47.55          1543320000
3           34.67          1543550400           45.03          1543510800
4           33.10          1543402800           42.18          1543420800

   apparentTemperatureMin  apparentTemperatureMinTime  apparentTemperatureMax  \
0                   33.73                  1545012000                   38.07
1                   36.20                  1543291200                   43.92
2                   31.04                  1543377600                   44.12
3                   30.30                  1543550400                   38.53
4                   29.11                  1543392000                   35.75

   apparentTemperatureMaxTime
0                  1544958000
1                  1543251600
2                  1543320000
3                  1543510800
4                  1543420800

[5 rows x 49 columns]
```

[117]:
```
y.head()
```

[117]:
```
0    0
1    0
2    0
3    0
4    0
Name: cab_type, dtype: int32
```

## 11 5.1 Escalado de los datos usando StandardScaler

StandardScaler calcula y resta la media de las muestras, para luego escalar la varianza de la unidad. La puntuación estándar (Z-Score) de una muestra $x$ se calcula como $z = \frac{x-\mu}{s}$ donde $\mu$ es la media de las muestras de entrenamiento o 0, si with_mean=False; y $s$ es la desviación estándar de las muestras de entrenamiento o 1, si with_std=False

[118]:
```
sc = StandardScaler()
ss_data = sc.fit_transform(X.values) # fit_transform recibe y regresa un ndarray
Xsc = pd.DataFrame(data=ss_data, columns=X.columns)
```

[119]:
```
Xsc.describe()
```

[119]:

| | day | month | source | destination | cab_type |
|---|---|---|---|---|---|
| count | 6.930710e+05 | 6.930710e+05 | 6.930710e+05 | 6.930710e+05 | 6.930710e+05 |
| mean | -5.163500e-15 | 2.777002e-14 | -3.688579e-15 | 3.895420e-15 | 4.205163e-14 |
| std | 1.000001e+00 | 1.000001e+00 | 1.000001e+00 | 1.000001e+00 | 1.000001e+00 |
| min | -1.682418e+00 | -1.191411e+00 | -1.593327e+00 | -1.593380e+00 | -1.120073e+00 |
| 25% | -4.802876e-01 | -1.191411e+00 | -7.237682e-01 | -7.238051e-01 | -1.120073e+00 |
| 50% | -7.957747e-02 | 8.393412e-01 | -1.440622e-01 | -1.440887e-01 | 8.927988e-01 |
| 75% | 1.022375e+00 | 8.393412e-01 | 7.254967e-01 | 7.254859e-01 | 8.927988e-01 |
| max | 1.222730e+00 | 8.393412e-01 | 1.595056e+00 | 1.595060e+00 | 8.927988e-01 |

| | name | price | distance | surge_multiplier |
|---|---|---|---|---|
| count | 6.930710e+05 | 6.930710e+05 | 6.930710e+05 | 6.930710e+05 |
| mean | -1.094983e-15 | 1.851081e-15 | 1.605019e-15 | -4.076322e-14 |
| std | 1.000001e+00 | 1.000001e+00 | 1.000001e+00 | 1.000001e+00 |
| min | -1.600311e+00 | -1.569978e+00 | -1.904786e+00 | -1.513489e-01 |
| 25% | -8.067826e-01 | -7.875113e-01 | -7.984905e-01 | -1.513489e-01 |
| 50% | -1.325389e-02 | -6.093468e-02 | -2.583969e-02 | -1.513489e-01 |
| 75% | 7.802749e-01 | 6.656419e-01 | 6.414497e-01 | -1.513489e-01 |
| max | 1.573804e+00 | 9.049218e+00 | 4.978831e+00 | 2.167290e+01 |

| | temperature | … | precipIntensityMax | uvIndexTime | temperatureMin |
|---|---|---|---|---|---|
| count | 6.930710e+05 | … | 6.930710e+05 | 6.930710e+05 | 6.930710e+05 |
| mean | -1.631159e-16 | … | 1.308501e-15 | 6.000910e-14 | -3.302525e-16 |
| std | 1.000001e+00 | … | 1.000001e+00 | 1.000001e+00 | 1.000001e+00 |
| min | -3.073765e+00 | … | -6.769040e-01 | -1.276567e+00 | -2.756637e+00 |
| 25% | -4.660053e-01 | … | -6.769040e-01 | -9.015686e-01 | -5.083753e-01 |
| 50% | 1.346418e-01 | … | -6.696595e-01 | -3.963619e-01 | 1.209524e-01 |
| 75% | 5.940476e-01 | … | 9.821031e-01 | 1.103633e+00 | 8.384168e-01 |
| max | 2.621975e+00 | … | 1.965554e+00 | 1.603632e+00 | 1.490938e+00 |

| | temperatureMinTime | temperatureMax | temperatureMaxTime |
|---|---|---|---|
| count | 6.930710e+05 | 6.930710e+05 | 6.930710e+05 |
| mean | 7.301407e-14 | 2.631438e-15 | 1.096761e-13 |
| std | 1.000001e+00 | 1.000001e+00 | 1.000001e+00 |
| min | -1.332391e+00 | -2.081705e+00 | -1.293805e+00 |
| 25% | -9.307654e-01 | -4.767570e-01 | -8.817118e-01 |
| 50% | -4.561169e-01 | -1.029776e-01 | -3.757237e-01 |
| 75% | 1.082579e+00 | 2.920593e-01 | 1.110942e+00 |
| max | 1.666762e+00 | 2.233586e+00 | 1.538685e+00 |

| | apparentTemperatureMin | apparentTemperatureMinTime |
|---|---|---|
| count | 6.930710e+05 | 6.930710e+05 |
| mean | -2.102476e-15 | -2.828620e-14 |
| std | 1.000001e+00 | 1.000001e+00 |
| min | -2.520361e+00 | -1.326621e+00 |
| 25% | -2.771964e-01 | -9.441913e-01 |
| 50% | 5.611406e-02 | -4.412704e-01 |

```
75%              8.408702e-01                 1.077970e+00
max              1.451236e+00                 1.580890e+00

       apparentTemperatureMax  apparentTemperatureMaxTime
count            6.930710e+05                6.930710e+05
mean             4.031099e-15               -3.731731e-14
std              1.000001e+00                1.000001e+00
min             -1.880878e+00               -1.246162e+00
25%             -7.823946e-01               -8.815137e-01
50%             -1.509828e-01               -3.762155e-01
75%              3.059979e-01                1.113633e+00
max              2.191584e+00                1.535583e+00

[8 rows x 49 columns]
```

## 12 5.2 Escalado de los datos usando MinMaxScaler

[120]:
```
scaler = MinMaxScaler()
norm_data = scaler.fit_transform(X.values)
Xnorm = pd.DataFrame(data=norm_data, columns=X.columns)
```

[121]:
```
Xnorm.head()
```

[121]:

```
        day  month    source  destination  cab_type      name     price  \
0  0.517241    1.0  0.454545     0.636364       0.0  0.583333  0.026316
1  0.896552    0.0  0.454545     0.636364       0.0  0.166667  0.089474
2  0.931034    0.0  0.454545     0.636364       0.0  0.416667  0.047368
3  1.000000    0.0  0.454545     0.636364       0.0  0.333333  0.247368
4  0.965517    0.0  0.454545     0.636364       0.0  0.500000  0.068421

   distance  surge_multiplier  temperature  …  precipIntensityMax  \
0  0.053571               0.0     0.611590  …            0.874572
1  0.053571               0.0     0.643957  …            0.891021
2  0.053571               0.0     0.506917  …            0.729267
3  0.053571               0.0     0.403811  …            0.000000
4  0.053571               0.0     0.483686  …            0.000685

   uvIndexTime  temperatureMin  temperatureMinTime  temperatureMax  \
0     0.913201        0.883145            0.913043        0.417488
1     0.045208        0.904987            0.053913        0.566092
2     0.088608        0.718238            0.123478        0.576355
3     0.173599        0.693120            0.206957        0.472906
4     0.130199        0.635967            0.135652        0.355911

   temperatureMaxTime  apparentTemperatureMin  apparentTemperatureMinTime  \
0            0.928177                0.776204                    0.938739
1            0.049724                0.863669                    0.077477
```

```
2            0.084715                0.680949                    0.120721
3            0.182320                0.654745                    0.207207
4            0.136280                0.612606                    0.127928

   apparentTemperatureMax  apparentTemperatureMaxTime
0                0.322832                    0.921348
1                0.529912                    0.033708
2                0.536991                    0.069288
3                0.339115                    0.168539
4                0.240708                    0.121723

[5 rows x 49 columns]
```

[122]:
```
Xnorm.describe()
```

[122]:
```
                day          month         source    destination  \
count  693071.000000  693071.000000  693071.000000  693071.000000
mean        0.579116       0.586684       0.499729       0.499736
std         0.344217       0.492429       0.313639       0.313633
min         0.000000       0.000000       0.000000       0.000000
25%         0.413793       0.000000       0.272727       0.272727
50%         0.551724       1.000000       0.454545       0.454545
75%         0.931034       1.000000       0.727273       0.727273
max         1.000000       1.000000       1.000000       1.000000

           cab_type           name          price       distance  \
count  693071.000000  693071.000000  693071.000000  693071.000000
mean        0.556455       0.504176       0.147843       0.276713
std         0.496803       0.315049       0.094169       0.145273
min         0.000000       0.000000       0.000000       0.000000
25%         0.000000       0.250000       0.073684       0.160714
50%         1.000000       0.500000       0.142105       0.272959
75%         1.000000       0.750000       0.210526       0.369898
max         1.000000       1.000000       1.000000       1.000000

       surge_multiplier    temperature  …  precipIntensityMax  \
count     693071.000000  693071.000000  …       693071.000000
mean           0.006935       0.539660  …            0.256165
std            0.045821       0.175570  …            0.378436
min            0.000000       0.000000  …            0.000000
25%            0.000000       0.457844  …            0.000000
50%            0.000000       0.563299  …            0.002742
75%            0.000000       0.643957  …            0.627827
max            1.000000       1.000000  …            1.000000

         uvIndexTime  temperatureMin  temperatureMinTime  temperatureMax  \
count  693071.000000   693071.000000       693071.000000   693071.000000
```

```
mean        0.443222        0.648991            0.444256        0.482402
std         0.347198        0.235429            0.333428        0.231734
min         0.000000        0.000000            0.000000        0.000000
25%         0.130199        0.529305            0.133913        0.371921
50%         0.305606        0.677466            0.292174        0.458539
75%         0.826401        0.846378            0.805217        0.550082
max         1.000000        1.000000            1.000000        1.000000

       temperatureMaxTime  apparentTemperatureMin  apparentTemperatureMinTime  \
count       693071.000000           693071.000000                693071.000000
mean             0.456773                0.634596                     0.456274
std              0.353047                0.251788                     0.343937
min              0.000000                0.000000                     0.000000
25%              0.145488                0.564802                     0.131532
50%              0.324125                0.648725                     0.304505
75%              0.848987                0.846317                     0.827027
max              1.000000                1.000000                     1.000000

       apparentTemperatureMax  apparentTemperatureMaxTime
count           693071.000000               693071.000000
mean                0.461853                    0.447979
std                 0.245552                    0.359487
min                 0.000000                    0.000000
25%                 0.269735                    0.131086
50%                 0.424779                    0.312734
75%                 0.536991                    0.848315
max                 1.000000                    1.000000

[8 rows x 49 columns]
```

## 13 6. Entrenamiento y evaluación del modelo de regresión logística

[123]:
```python
from sklearn.metrics import accuracy_score, confusion_matrix, precision_score,␣
↪recall_score, f1_score

def regression(X, y):
    # Escalar los datos
    scaler = StandardScaler()
    X_scaled = scaler.fit_transform(X)

    # Dividir los datos en el conjunto de entrenamiento y el de prueba
    X_train, X_test, y_train, y_test = train_test_split(X_scaled, y,␣
↪test_size=0.3, shuffle=True)

    # Construir el modelo
```

```
    model = LogisticRegression()
    model.fit(X_train, y_train)

    # Realizar las predicciones
    y_pred = model.predict(X_test)

    # Evaluar el modelo
    accuracy = accuracy_score(y_test, y_pred)
    conf_matrix = confusion_matrix(y_test, y_pred)
    precision = precision_score(y_test, y_pred)
    recall = recall_score(y_test, y_pred)
    f1 = f1_score(y_test, y_pred)

    # Mostrar los resultados
    print("Accuracy:", accuracy)
    print("Confusion Matrix:")
    print(conf_matrix)
    print("Precision:", precision)
    print("Recall (Sensibilidad):", recall)
    print("F1 Score:", f1)

    return y_test, y_pred

# Llamar a la función regresión con datos X e y
y_test, y_pred = regression(X, y)
```

```
Accuracy: 1.0
Confusion Matrix:
[[ 92130      0]
 [     0 115792]]
Precision: 1.0
Recall (Sensibilidad): 1.0
F1 Score: 1.0
```

# 14 7.1 Probar la regresión con los datos estandarizados (Standard-Scaler)

[124]:
```
y_test_sc, y_sc = regression(Xsc, y)
plt.show()
```

```
Accuracy: 1.0
Confusion Matrix:
[[ 92324      0]
 [     0 115598]]
Precision: 1.0
Recall (Sensibilidad): 1.0
F1 Score: 1.0
```

## 15 7.2 Probar la regresión con los datos normalizados (MinMaxScaler)

[125]:
```
y_test_norm, y_norm = regression(Xnorm, y)
```

```
Accuracy: 1.0
Confusion Matrix:
[[ 92286      0]
 [     0 115636]]
Precision: 1.0
Recall (Sensibilidad): 1.0
F1 Score: 1.0
```

## 16 Conclusión

Accuracy (Precisión): Significa que el porcentaje de predicciones correctas de todas las predicciones realizadas es del 100%. Todas las muestras en el conjunto de prueba se clasificaron correctamente.

Confusion Matrix: Muestra que no hubo falsos positivos en el conjunto de prueba. Todos los datos negativos fueron correctamente identificados como negativos, y todos los datos positivos fueron correctamente identificados como positivos.

Recall: Significa que el porcentaje de muestras positivas que fueron correctamente identificadas por el modelo es del 100%. No hubo falsos negativos en las predicciones

F1 Score: El puntaje de F1 indica que no hubo errores ni falsos positivos ni falsos negativos en las predicciones